12TH BIOLOGY · CHAPTER 22 · SOLVED EXERCISE

Biostatistics

Complete textbook exercise: MCQs, Short Questions, Long Questions and Inquisitive Questions

SECTION 1: MULTIPLE CHOICE QUESTIONS

1. Which of the following is a primary data collection method?

(a) Government reports
(b) Published articles
(c) Experiment and surveys
(d) Online databases

Answer: (c) Experiment and surveys — Primary data are collected directly; the other three are existing (secondary) sources.

2. What does mode represent in a dataset?

(a) The average of values
(b) The value occurring most frequently
(c) The difference between two values
(d) None of these

Answer: (b) The value occurring most frequently — The mode is the value that appears most frequently.

3. Which statistical measure is also called the 50th percentile?

(a) Mean
(b) Median
(c) Mode
(d) Range

Answer: (b) Median — The median is the value below which 50% of the data points lie.

4. If the maximum value in a dataset is 50 and the minimum value is 20, what is the range?

(a) 30
(b) 25
(c) 20
(d) 115

Answer: (a) 30 — Range = maximum – minimum = 50 – 20 = 30.

5. What does it mean if the 40th percentile value is 148cm?

(a) 40% of values are above 148cm
(b) 40% of values are below or equal to 148cm
(c) All the values are equal to 148cm
(d) 60% of values are above 148cm

Answer: (b) 40% of values are below or equal to 148cm — A percentile is the value below which the given percentage of observations falls. (Option (d) follows from it, but (b) is the definition.)

studysapiens.com

6. In a pie chart, what does the full circle represent?

(a) 180° (b) 280°
(c) 380° (d) 360°

Answer: (d) 360° — The entire circle corresponds to 100% or 360°.

7. What do error bars represents in a chart?

(a) Average (b) Variability or uncertainty in data
(c) Both of these (d) None of these

Answer: (b) Variability or uncertainty in data — The bar itself shows the average (mean); the error bar shows the variability.

8. In an experiment, what does the dependent variable represent?

(a) The factor kept constant (b) The factor being changed
(c) The factor ignored (d) All of these

Answer: None of the printed options is correct. The dependent variable is the factor that is **measured or observed**; it changes in response to the independent variable.

Textbook error: (a) describes the control variable and (b) describes the independent variable. If the examiner insists on one option, (b) is the nearest, because the dependent variable is the factor that changes in response; but this is not a correct statement of it.

9. What is the main purpose of including a control group in an experiment?

(a) To test the dependent variable
(b) To provide a baseline for comparison
(c) To collect random data
(d) All of these

Answer: (b) To provide a baseline for comparison — The control group does not receive the treatment, so the experimental group is compared with it.

10. Which group in an experiment receives the treatment?

(a) Control group (b) Dependent group
(c) Experimental group (d) Independent group

Answer: (c) Experimental group — The experimental group is exposed to the treatment or change being tested.

SECTION 2: SHORT QUESTIONS

1. Explain why biostatistics is important in medical research?

Biostatistics provides the framework for designing studies and analysing data, so that the outcomes are valid and reliable. With statistical methods, the safety and effectiveness of new drugs, treatments and medical devices are evaluated. This lets healthcare professionals make evidence-based decisions that improve patient care and advance medical knowledge.

2. Describe the difference between descriptive and inferential statistics?

Descriptive statistics	Inferential statistics
Describe the characteristics of the collected data.	Allow conclusions about populations from sample data.
Examples: mean, median, mode, standard deviation.	Examples: hypothesis testing, regression analysis, modelling.

3. What is the relationship between percentiles and quartiles?

Quartiles are particular percentiles; they divide the ordered data into four equal parts. The 25th percentile is the first quartile (Q1), the 50th percentile is the median (the second quartile), and the 75th percentile is the third quartile (Q3).

4. How are independent and dependent variables related in experimental design?

The independent variable is the factor that the experimenter changes. The dependent variable is the factor that is measured; it changes in response to the independent variable. So the independent variable is the cause and the dependent variable is the effect. Example: type of diet (independent) and weight gain of mice (dependent).

5. The height of six children are 110, 115, 118, 120, 122, and 125. What is the median height of the children?

The data are already in ascending order: 110, 115, **118, 120**, 122, 125. Here n = 6 (even), so the median is the average of the 3rd and 4th values.

$$\text{Median} = (118 + 120) / 2 = 238 / 2 = 119$$

The median height of the children is **119** (the unit is not given in the question; it would be cm).

6. What does a legend in a chart present?

A legend is a key or guide on a chart, graph or map. It explains the meaning of the symbols, colours, patterns or labels used in it. For example, in a bar chart with a different colour for each category, the legend tells which category each colour represents.

7. What are error bars?

Error bars are graphical elements placed on bar charts (and other plots) to show the variability or uncertainty in data. They show how much the individual data points deviate from the central value (e.g., the mean) represented by the bar. The two common types are range error bars and standard deviation error bars.

SECTION 3: LONG QUESTIONS

1. Explain the concepts of mean, median, and mode as measures of central tendency. Illustrate with examples when each measure would be most appropriate.

1. Mean

The **mean** (average) is a measure of central tendency that gives the central value of a dataset. In biology it summarizes large datasets with one typical value, such as average size or growth rate.

Ungrouped data: Mean = $\Sigma x / n$ (Σx = sum of all values, n = number of values)

Grouped data: Mean = $\Sigma f_i x_i / \Sigma f_i$ (x_i = class midpoint, f_i = class frequency)

Example (ungrouped): Weights of five birds: 400 g, 420 g, 470 g, 390 g and 460 g.

$\Sigma x = 400 + 420 + 470 + 390 + 460 = 2140$ g; $n = 5$

Mean = $2140 / 5 = 428$ g

2. Median

The **median** is a measure of central tendency that represents the middle value of a dataset when the data are arranged in order. It represents the centre better when the data are not normally distributed, e.g., population size, disease prevalence or body measurements.

Steps: (1) Arrange the data in ascending or descending order. (2) Count the number of values, n. (3) If n is odd, the median is the value at position $(n + 1)/2$. (4) If n is even, the median is the average of the values at positions $n/2$ and $(n/2) + 1$.

Example (odd n): Heights of five plants: 50, 65, 60, 75, 80 cm. In order: 50, 60, **65**, 75, 80. Position = $(5 + 1)/2$ = 3rd value, so the median = **65 cm**.

Example (even n): Weights of six animals: 300, 400, 330, 315, 450, 500 kg. In order: 300, 315, **330, 400**, 450, 500. Median = $(330 + 400)/2$ = **365 kg**.

3. Mode

The **mode** is the value (or values) that appears most frequently in a dataset. In biology it shows the most common trait in a sample, e.g., the most frequent number of flowers on a plant or the most common seed size. For ungrouped data there is no formula; for grouped data:

$$\text{Mode} = l + [\ (f_m - f_1) / (2f_m - f_1 - f_2)\ ] \times h$$

where l = lower boundary of the modal class, f_m = frequency of the modal class, f_1 = frequency of the class before it, f_2 = frequency of the class after it, and h = class width.

Example (ungrouped): Heights of ten plants in order: 50, 50, 50, 50, 55, 55, 60, 60, 65, 70 cm. The value 50 appears 4 times, the most. Mode = **50 cm**.

Example (grouped): Weights of zoo animals: 10–20 (5), 20–30 (12), 30–40 (18), 40–50 (10), 50–60 (3). The modal class is 30–40 (highest frequency 18). So l = 30, f_m = 18, f_1 = 12, f_2 = 10, h = 10.

$$\text{Mode} = 30 + [\ (18 - 12) / (36 - 12 - 10)\ ] \times 10 = 30 + (6/14) \times 10 = 34.29 \text{ kg}$$

When each measure is most appropriate

Measure	Most appropriate when	Example
Mean	A large dataset is to be summarized by one typical value	Average weight of birds; average growth rate of plants
Median	The data are not normally distributed	Population size, disease prevalence, body measurements
Mode	The most common trait or value is needed	Most frequent number of flowers on a plant; most common seed size

2. Describe standard deviation, range, and percentiles as measures of variability. Compare their advantages and limitations in interpreting biological data?

1. Standard deviation

The **standard deviation** measures how dispersed the data are in relation to the mean. A small standard deviation means the data are clustered tightly around the mean; a large one means they are more spread out. In biology it shows the variability in measurements such as growth rates of plants or enzyme concentrations.

$$\sigma = \sqrt{[\ \Sigma(x_i - \mu)^2 / N\ ]}$$

where σ = standard deviation, x_i = each data point, μ = mean and N = total number of data points.

Steps: (1) Find the mean. (2) Subtract the mean from each data point. (3) Square each deviation. (4) Find the average of the squared deviations (variance). (5) Take its square root.

Example: Heights of 9 plants: 56, 65, 74, 75, 76, 77, 80, 81, 91 cm. Mean μ = 675/9 = 75 cm.

x_i	56	65	74	75	76	77	80	81	91
$x_i - \mu$	–19	–10	–1	0	1	2	5	6	16
$(x_i - \mu)^2$	361	100	1	0	1	4	25	36	256

$$\Sigma(x_i - \mu)^2 = 784;\ \text{Variance} = 784 / 9 = 87.11$$

$$\sigma = \sqrt{87.11} = 9.33 \text{ cm}$$

2. Range

The **range** is the simplest measure of the spread of data. It is the difference between the maximum and minimum values. A larger range means the data points are more dispersed; a smaller range means they are closely clustered.

$$\text{Range} = X_{max} - X_{min}$$

Example: Ages of a group: 25, 20, 22, 30, 42, 45, 50. In order: 20, 22, 25, 30, 42, 45, 50. Greatest value = 50, smallest value = 20. Range = 50 – 20 = **30**.

3. Percentiles

A **percentile** is the value in a dataset below which a given percentage of observations falls. The 25th percentile is the first quartile (Q1), the 50th percentile is the median, and the 75th percentile is the third quartile (Q3). Percentiles help to understand the distribution of biological data, whether skewed or normal.

$$R = (P / 100) \times (n + 1)$$ (R = rank, P = desired percentile, n = number of observations)

If R is an integer, the value at the Rth position of the ordered data is the percentile. If R is not an integer, interpolate between the two nearest ranks.

Example: Find the 40th percentile of the heights 160, 162, 168, 165, 170, 175, 172, 178, 180, 182 cm. In order: 160, 162, 165, 168, 170, 172, 175, 178, 180, 182; n = 10.

$$R = (40/100) \times (10 + 1) = 4.4$$ (between the 4th value 168 and the 5th value 170)

$$\text{40th percentile} = 168 + 0.4 \times (170 - 168) = 168.8 \text{ cm}$$

So 40% of the observations fall below or equal to 168.8 cm.

Comparison

Measure	Advantages	Limitations
Standard deviation	Uses every data point; shows how tightly the data cluster around the mean	Longer calculation; affected by extreme values
Range	The simplest measure; quick to calculate	Uses only the two extreme values; says nothing about the values in between
Percentiles	Show the position of a value in the distribution; useful for skewed data as well as normal data	Do not give a single figure for the spread; may need interpolation

The textbook describes the three measures but does not list their advantages and limitations; the comparison table is reasoned from the textbook definitions.

3. What are error bars? Explain their types (range error bars, standard deviation error bars) and significance in experimental biology. Illustrate your answer with a hypothetical dataset?

Error bars are graphical elements placed on bar charts (and other plots) to show the variability or uncertainty in data. They show how much the individual data points deviate from the central value (e.g., the mean) represented by the bar.

Range error bars use the smallest and greatest data points of each group; the bar extends from the minimum to the maximum value and shows the full spread of the data.

Standard deviation (SD) error bars show the average distance of the data points from the mean; they extend above and below the mean by one or more standard deviations (± SD).

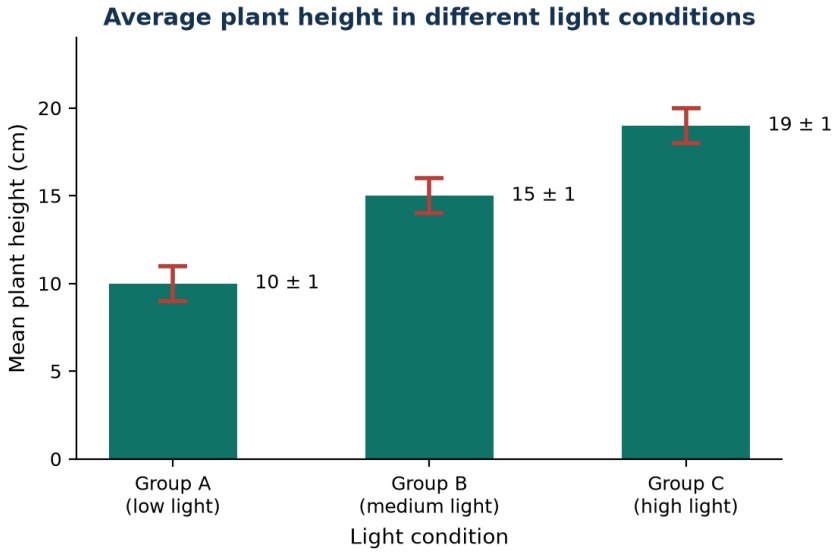
Significance in experimental biology: A bar shows only the mean of a group. Error bars show how variable the measurements are and so how reliable the mean is. If the error bars of two groups do not overlap, the difference between the groups is more convincing; if they overlap, the difference may not be real.

The overlap rule is a general guide and is not stated in the textbook.

Example: Heights (cm) of bean plants after four weeks in three light conditions.

Group	Plant 1	Plant 2	Plant 3	Mean	Range (max – min)	Range error (±)	SD (±)
A (low light)	10	11	9	10	2	1	1
B (medium light)	14	16	15	15	2	1	1
C (high light)	18	20	19	19	2	1	1

1. **Calculate the mean** of each group; draw the bars at heights 10, 15 and 19.
2. **Calculate the spread:** SD ≈ 1 for each group; range: A 9–11, B 14–16, C 18–20.
3. **Draw the error bars:** SD bars from mean – 1 to mean + 1 (A 9–11, B 14–16, C 18–20); range bars from minimum to maximum. Add a horizontal cap at each end.



Bar chart with error bars (here the SD bars and the range bars are equal)

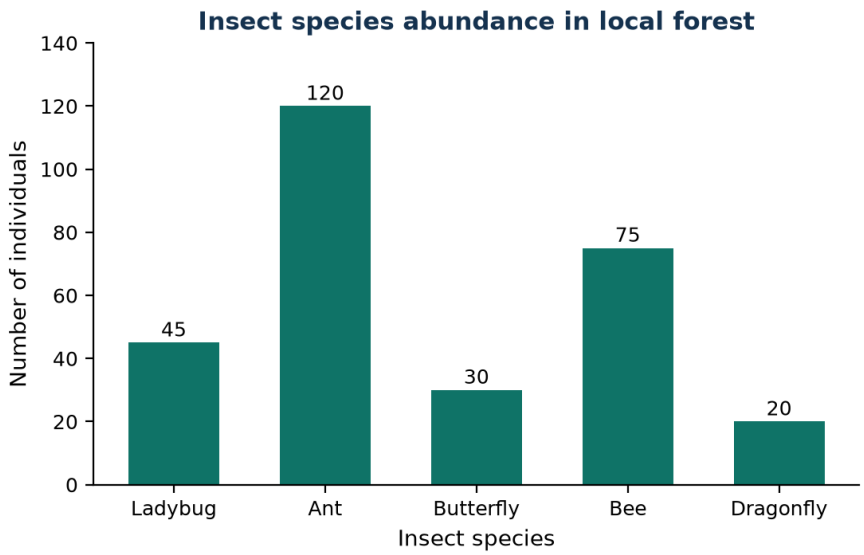
4. Compare and construct bar charts and pie charts. Which situations favour one over the other? Support your answer with real-world biological or health data examples.

Bar chart

A **bar chart** (bar graph) displays categorical data with rectangular bars; the length or height of each bar is proportional to the value it represents. It is used to compare quantities across distinct categories, e.g., counts of different species or results of experimental groups. Categories are on the x-axis and values on the y-axis; bars may be vertical or horizontal.

1. **Purpose and data:** identify the categorical variable (insect species) and the numerical variable (number counted).
2. **Determine the axes:** x-axis = insect species; y-axis = number of individuals.
3. **Scale the numerical axis:** the greatest value is 120, so use 0 to 140 with an increment of 20.
4. **Draw the bars:** ladybug 45, ant 120, butterfly 30, bee 75, dragonfly 20; bars of equal width, evenly spaced.
5. **Add labels and title:** axis labels and the chart title.

studysapiens.com

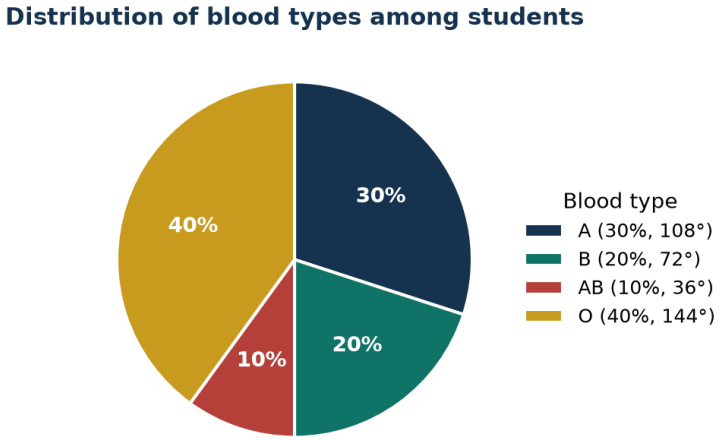


Bar chart: insect species counted in a local forest (textbook data)

Pie chart

A **pie chart** is a circular graph divided into sectors (slices), each representing a proportion of the whole dataset. The whole circle is 100% (360°), and the angle of each sector is proportional to its category's share. It is ideal for showing categories as portions of a whole, e.g., types of cells, species distribution or survey responses.

1. **Organize the data** in a table with the frequency of each category.
2. **Add up** all frequencies: 30 + 20 + 10 + 40 = 100 students.
3. **Calculate the sector angles:** sector angle = (category value / total) × 360°. A = 108°, B = 72°, AB = 36°, O = 144°.
4. **Draw** a circle; from a reference radius, mark each angle in sequence with a protractor; use different colours.
5. **Label** each sector, add a legend if needed and a descriptive title.



Pie chart: blood types of 100 students (textbook data)

Comparison

Bar chart	Pie chart
Rectangular bars; height is proportional to the value.	Circle divided into sectors; angle is proportional to the share.
Compares quantities across distinct categories.	Shows categories as portions of one whole (100%).
Has an x-axis and a y-axis with a scale; exact values can be read; error bars can be added.	Has no axes; shows percentages only.
Favoured for counts or measurements of separate groups, e.g., number of individuals of five insect species; mean plant height in three light conditions.	Favoured when the parts add up to a whole, e.g., blood types of 100 students; types of cells in a blood sample.

5. Discuss different types of data presentation tools in biostatistics (bar chart, pie chart and error bars). For each, explain an appropriate biological example where it would be the best choice.

Tool	What it shows	Best biological example
Bar chart	Categorical data as rectangular bars whose height is proportional to the value; compares quantities across distinct categories	Number of individuals of different insect species counted in a forest (ladybug 45, ant 120, butterfly 30, bee 75, dragonfly 20)
Pie chart	A circle (360° = 100%) divided into sectors; each sector is a proportion of the whole	Blood types in a class of 100 students (A 30%, B 20%, AB 10%, O 40%)
Error bars	Variability or uncertainty of the data around the mean of each bar; drawn from the range or the standard deviation	Mean height of bean plants grown in low, medium and high light (10 ± 1, 15 ± 1 and 19 ± 1 cm)

Bar chart: the best choice when separate groups are compared, because the heights of the bars can be compared directly on a scaled axis.

Pie chart: the best choice when the categories together make one whole, because each slice shows the share of its category at a glance.

Error bars: the best choice when each bar is a mean of repeated measurements, because they show how far the individual values spread around that mean. The charts for these examples are drawn in Long Questions 3 and 4.

INQUISITIVE QUESTIONS

1. A pharmaceutical company claims their drug improves recovery time from flu. Propose an experimental design using biostatistics principles, and explain how you would ensure validity and reliability of results?

Step	Design
Question	Does the drug reduce the recovery time from flu?
Hypothesis	Flu patients given the drug will recover in fewer days than patients not given the drug.
Variables	Independent: treatment (drug or no drug). Dependent: recovery time in days. Control: age group, severity of flu, time since symptoms began, diet, rest and other medicines.
Groups	Experimental group: patients given the drug. Control group: similar patients given a dummy tablet (placebo).
Procedure	Select a large sample of flu patients; assign them randomly to the two groups; treat both groups in the same way; record the day of recovery of every patient.
Analysis	Calculate the mean (and median) recovery time and the standard deviation of each group; draw a bar chart with error bars; apply a statistical test.
Conclusion	If the drug group recovers significantly faster, the hypothesis is supported; otherwise it is rejected.

Ensuring validity: a control group as baseline, random assignment, all control variables kept the same, and an adequate sample size, so that only the drug can explain any difference. **Ensuring reliability:** a large sample, the same method of measurement for every patient, repetition of the trial, and transparent reporting so that others can reproduce it.

The use of a placebo and blinding is not mentioned in the textbook; it is standard practice in drug trials.

2. Evaluate the usefulness of range versus standard deviation in representing variability. Which one would you trust more in medical research and why?

The **range** is the simplest measure and is quick to find, but it uses only the maximum and minimum values. One unusual patient can change it greatly, and it tells nothing about how the other values are spread. The **standard deviation** uses every data point and shows how tightly the data cluster around the mean.

In medical research the **standard deviation** is more trustworthy, because conclusions about a drug or disease must reflect all patients, not just the two extreme ones. The range is useful only as a quick first look at the spread.

3. During an epidemic, mortality rate estimates vary depending on whether the mean, median, or percentile is used. Discuss how policymakers might misinterpret such statistics and how biostatistics can prevent this.

How misinterpretation occurs: Epidemic data are usually not normally distributed; a few areas or age groups may have very high death rates. The mean is pulled up by these extreme values, so a policymaker who sees only the mean may overestimate the danger for a typical area. The median shows the middle value, so it may hide the badly affected areas. A percentile describes only one point of the distribution. Using one figure alone, or comparing a mean from one report with a median from another, leads to wrong decisions about resources.

How biostatistics prevents it:

- Reporting the measure used, together with a measure of spread (standard deviation, range or percentiles).
- Choosing the median for skewed data and stating why.
- Analysing at-risk groups separately instead of giving one overall figure.
- Interpreting results with their context, limitations and assumptions, and presenting them with clear charts and transparent methods.

studysapiens.com